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README

3c.CREATION FOR FILE TRANSFER USING TCP SOCKETS

AIM

To write a python program for creating File Transfer using TCP Sockets Links

ALGORITHM:

1. Import the necessary python modules.
2. Create a socket connection using socket module.
3. Send the message to write into the file to the client file.
4. Open the file and then send it to the client in byte format.
5. In the client side receive the file from server and then write the content into it.

PROGRAM

Server.py

```
import socket
import os
SERVER_HOST = "127.0.0.1"
SERVER_PORT = 5001
BUFFER_SIZE = 4096
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((SERVER_HOST, SERVER_PORT))
server.listen(1)
print("Server listening on port", SERVER_PORT)
conn, addr = server.accept()
print("Connected from:", addr)
try:
    filename = conn.recv(BUFFER_SIZE).decode()
    filesize = int(conn.recv(BUFFER_SIZE).decode())
    print(f"Receiving file: {filename}")
    print(f"File size: {filesize} bytes")
    received_size = 0
```

```
    with open("received_" + filename, "wb") as f:
        while received_size < filesize:
            bytes_read = conn.recv(BUFFER_SIZE)
            if not bytes_read:
                break
            f.write(bytes_read)
            received_size += len(bytes_read)
            progress = (received_size / filesize) * 100
            print(f"Progress: {progress:.2f}%", end="\r")
    print("\nFile received successfully!")
except Exception as e:
    print("Error:", e)
finally:
    conn.close()
    server.close()
```

client.py

```
import socket
import os
SERVER_HOST = "127.0.0.1"
SERVER_PORT = 5001
BUFFER_SIZE = 4096
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_HOST, SERVER_PORT))
try:
    filename = input("Enter file name to send: ")
    if not os.path.exists(filename):
        print("File does not exist!")
        client.close()
        exit()
    filesize = os.path.getsize(filename)
    client.send(filename.encode())
    client.send(str(filesize).encode())
    print(f"Sending {filename} ({filesize} bytes)")
    sent_size = 0
    with open(filename, "rb") as f:
        while True:
            bytes_read = f.read(BUFFER_SIZE)
            if not bytes_read:
                break
            client.sendall(bytes_read)
            sent_size += len(bytes_read)
            progress = (sent_size / filesize) * 100
            print(f"Progress: {progress:.2f}%", end="\r")
    print("\nFile sent successfully!")
except Exception as e:
    print("Error:", e)
finally:
    client.close()
```

OUPUT

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS C:\Users\acer\Computer Network ex8> python server.py
PS C:\Users\acer\Computer Network ex8> python server.py
Server listening on port 5001
Connected from: ('127.0.0.1', 56671)
Receiving file: client.py
File size: 1031 bytes
Progress: 100.00%
File received successfully!
PS C:\Users\acer\Computer Network ex8>
```

```
PS C:\Users\acer\Computer Network ex8> python client.py
Enter file name to send: client.py
Sending client.py (1031 bytes)
Progress: 100.00%
File sent successfully!
PS C:\Users\acer\Computer Network ex8>
```

```
powershell
powershell
```

RESULT

Thus, the python program for creating File Transfer using TCP Sockets Links was successfully created and executed.